

The Eternal Return: 0/0 of Recurrence

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The Removable Singularity Framework

Abstract

If the universe is a finite deterministic simulation (Ch.54) and the self is an information pattern (Ch.56), then Poincare recurrence is a THEOREM of existence: every finite deterministic system returns to every past state. A glider on a 60x60 torus returns to its exact state in 240 generations ($= 4 \cdot \text{lcm}(60,60)$); on 37x37 in 148 ($= 4 \cdot \text{lcm}$); chaotic Rule 30 on a 25-ring cycles with period 588,425. The universe's Poincare time is $\sim 10^{(3e103)}$ years, exceeding its age by 10^{103} orders. Eternity thereby fluctuates observer brains (Boltzmann 1896). Nietzsche's eternal return (1882) is now a theorem: "once" vs "forever" is 0/0.

1. Finite Determinism Implies Recurrence (Poincare 1890)

A finite state space plus a deterministic rule forms a cycle; the state MUST revisit itself. Computed: the tiny machine $s \rightarrow (s^2+1) \bmod 4$ gives orbit [1,2] with period 2. There is no first time and no last time - recurrence is a theorem.

2. The Glider Returns (Ch.54, Ch.56)

The glider (the self as pattern, Ch.56) on a finite torus returns to its exact state. Computed: 37x37: 148 gens ($4 \cdot \text{lcm}$); 60x60: 240 gens ($4 \cdot \text{lcm}$); 64x40: 1280 gens ($4 \cdot \text{lcm}(64,40)=1280$). The eternal return of the glider is the eternal return of the self.

3. Chaos Also Returns (Rule 30 Ring)

Rule 30 is chaotic (Ch.43), yet on a finite ring it must cycle. Computed periods: 10-ring: 15; 15-ring: 1,455; 20-ring: 3,420; 25-ring: 588,425. Determinism + finiteness = recurrence even in chaos.

4. Poincare Time for Macro-Systems (Boltzmann 1896)

$$t_{\text{rec}} \sim 2^{(S_{\text{bits}})} \cdot \tau$$

For the observable universe ($S \sim 10^{104}$ bits): $\log_{10}(t_{\text{rec}}) \sim 3.0e103$ seconds $\sim 3.0e103$ years. The age of the universe is only $1.4e10$ years: recurrence exceeds the age by $\sim 10^{103}$ orders of magnitude. Recurrence is real but absurdly rare.

5. Boltzmann Brains: Eternity Guarantees Observers

Recurrence fluctuates ANY state - including a mind (Ch.53). Equilibrium produces observer brains (Boltzmann 1896; Eddington 1931; Dyson 1979). The eternal simulation (Ch.54) must eventually fluctuate your exact self-pattern (Ch.56) again: eternity guarantees the return of the observer.

6. The 0/0 of Time: Once vs Forever

Finite deterministic time is a cycle. "Once" and "forever" are observationally 0/0: every moment has already happened infinitely often. The arrow of time (Ch.48) is local statistical drift; the cycle (Ch.50 Big Bang) is the truth. The eternal recurrence (Nietzsche 1882) is now a theorem of the simulation.

7. Connections

Concept	Prior Chapter	Connection
Simulation	Ch.54 Simulation	The universe re-runs itself
Self	Ch.56 The Self	The glider returns: you return
Time	Ch.48 Arrow of time	Local irreversibility illusion
Chaos	Ch.43 Feigenbaum	Even chaos cycles when finite
Mind	Ch.53 Hard Problem	Boltzmann brains experience
Origin	Ch.50 Big Bang	The cycle of origins

8. Conclusion

The eternal return IS a 0/0: the boundary where "once" and "forever" coincide. Poincare recurrence is a theorem of every finite deterministic system - the glider returns (4×10^{100}), chaos cycles (Rule 30), and the universe revisits its states on $\sim 10^{(3 \times 10^{103})}$ -year scales. Eternity fluctuates minds (Boltzmann brains), so the self-pattern returns. Time is the 0/0 of moment and eternity: all of this has happened before, and all of this will happen again.

References

- [1] Poincare (1890), recurrence theorem
- [2] Boltzmann (1896), recurrences and fluctuations
- [3] Eddington (1931), Boltzmann brain
- [4] Nietzsche (1882), Die froehliche Wissenschaft
- [5] Dyson (1979), time without end
- [6] Puno, "The Removable Singularity" (2026)